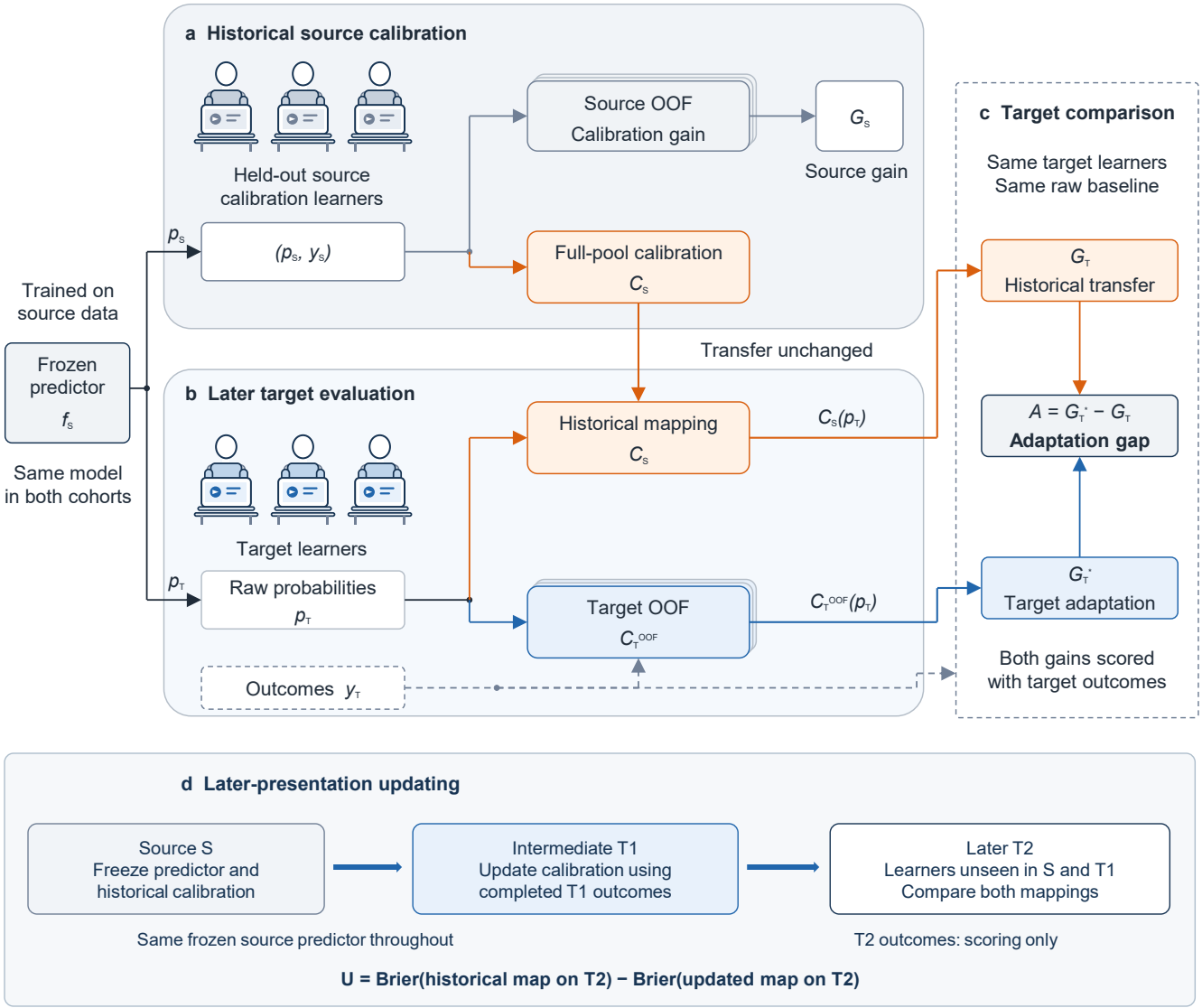

a Historical source calibration
Held-out source calibration learners
p_S
(p_S, y_S)
Source OOF Calibration gain
G_S
Source gain
Full-pool calibration C_S
Trained on source data
Frozen predictor f_S
Same model in both cohorts
Transfer unchanged
b Later target evaluation
Historical mapping C_S
$C_S(p_T)$
Target learners
p_T
Raw probabilities p_T
Target OOF C_T^{OOF}
$C_T^{OOF}(p_T)$
Outcomes y_T
c Target comparison
Same target learners
Same raw baseline
G_T
Historical transfer
$A = G_T^* - G_T$
Adaptation gap
G_T^*
Target adaptation
Both gains scored with target outcomes
d Later-presentation updating
Source S
Freeze predictor and historical calibration
Intermediate T1
Update calibration using completed T1 outcomes
Later T2
Learners unseen in S and T1
Compare both mappings
Same frozen source predictor throughout
T2 outcomes: scoring only
U = Brier(historical map on T2) − Brier(updated map on T2)